

SKS Technologies — AI-Assisted Quoting Tool

Full Product Framework & Development Handoff Document

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1. Product Vision

A purpose-built quoting tool for SKS Technologies that combines a live supplier product database with AI-assisted scope extraction, structured area/space building, and dual-mode output (Quick Quote for small jobs, Large Quote for full project submissions). The tool reduces quote build time, improves margin consistency, and produces professional client-ready documents.

2. Quote Modes

The user selects a mode at the start of every new quote. Mode determines the UI flow, complexity of output, and document format.

Quick Quote

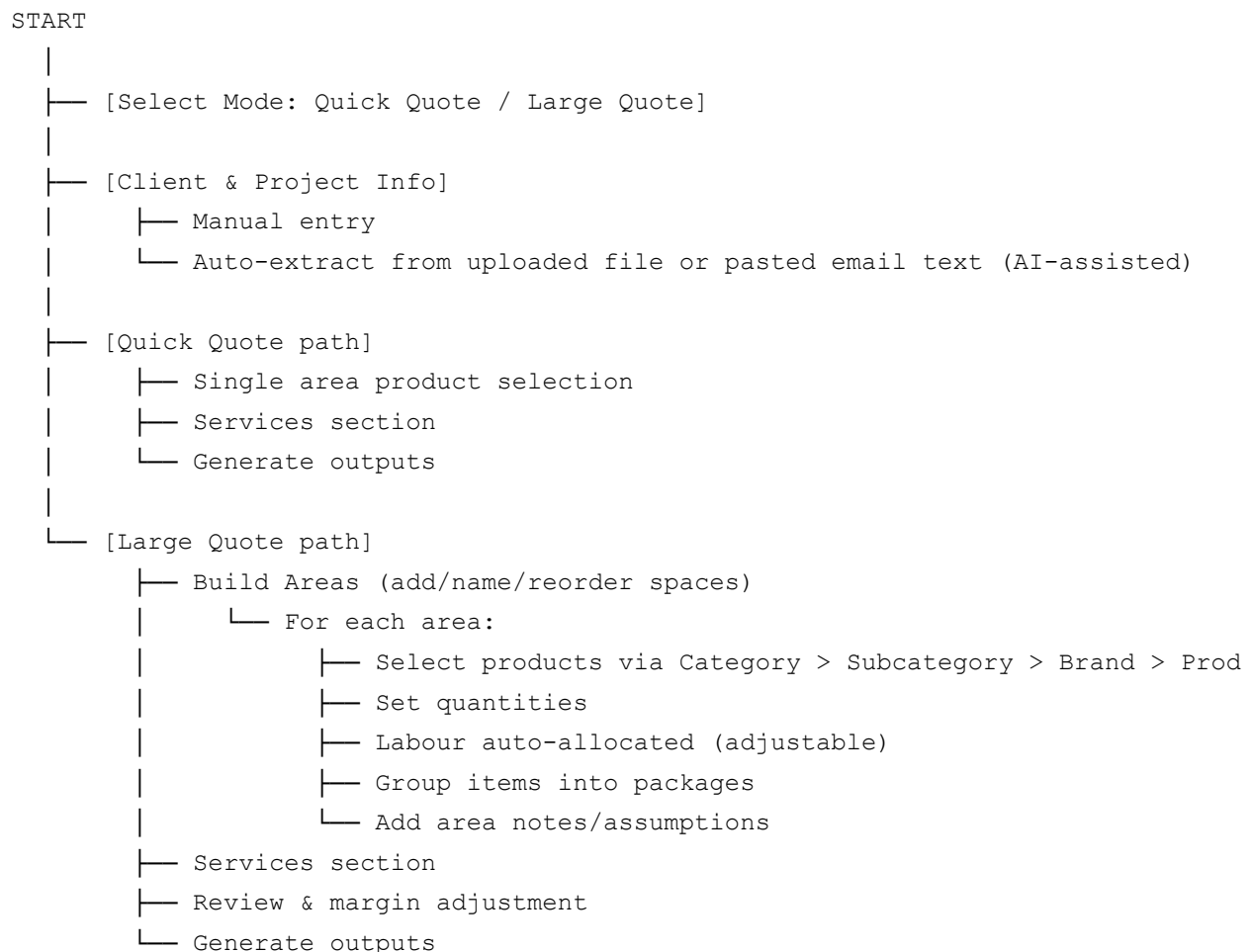
- **For:** Small jobs, minor variations, supply-only or single-trade work
- **Single area only** — no multi-space builder
- **Simplified client fields** (name, address, contact)
- **Fast product selection** — category dropdown, quantity, done
- **Output:** Single-page quote document + basic cost summary
- **Typical use:** One-room installs, single product supply, cabling runs, minor additions

Large Quote

- **For:** Full project submissions, tender responses, multi-room/multi-trade work
- **Multi-area builder** — user constructs each space individually
- **Full client + project fields**
- **Package grouping, area-level notes, per-area summaries**
- **Services section** with full breakdown
- **Output:** Full proposal document with cover page, scope summary, area breakdowns, exclusions, terms
- **Typical use:** Commercial fitouts, education, hospitality, government tenders

The quote mode can be upgraded mid-session (Quick → Large) without losing entered data.

3. Application Flow



4. Database Architecture

The tool is built on a structured internal database that stores all supplier data, products, labour rates, and configuration. This is the backbone of the tool.

4.1 Supplier Database

Each supplier has a profile and associated price sheet(s).

Supplier record:

Field	Type	Notes
supplier_id	string	Auto-generated
supplier_name	string	e.g. Amber Technology, Rutledge, Hills
supplier_code	string	Short code for internal use
contact_name	string	Account manager
contact_email	string	
contact_phone	string	
lead_time_days	integer	Default lead time
preferred	boolean	Flag as preferred supplier
notes	string	Internal notes
last_updated	date	When price list was last loaded

Price sheet upload process:

1. Admin uploads supplier price list (CSV or XLSX)
2. Tool maps columns to internal schema (column mapper UI)
3. Existing products for that supplier are updated; new ones are added
4. Price history is retained (previous cost price archived with date)
5. Admin reviews changes before committing

Supplier price sheet schema (normalised internal format):

Field	Type	Notes
product_id	string	Internal unique ID
supplier_id	string	FK to Supplier
supplier_part_no	string	Supplier's own part number
manufacturer	string	Brand name (e.g. Biamp, Crestron, Samsung)
manufacturer_part_no	string	OEM part number
description	string	Full product description
short_description	string	Display name in dropdowns
category	string	e.g. Audio, Visual, Control, Networking, Cabling
subcategory	string	e.g. Speakers, DSP, Displays, Switchers
unit	string	each / m / lot
cost_price	decimal	Ex-GST buy price
rrp	decimal	RRP if available
uom	string	Unit of measure
is_active	boolean	Hide discontinued products
notes	string	
last_updated	date	

4.2 Category & Subcategory Structure

The category tree is maintained in the database and controls the dropdown hierarchy.

Suggested base category/subcategory structure (AV/data/electrical):

Audio

- └─ Ceiling Speakers
- └─ Surface Mount Speakers
- └─ Subwoofers

- |— DSP / Audio Processors
- |— Amplifiers
- |— Microphones (ceiling array)
- |— Microphones (handheld / lapel)
- |— Audio Conferencing
- └─ Racks & Accessories

Visual

- |— Displays (commercial)
- |— Displays (interactive / touch)
- |— Video Walls
- |— Projectors
- |— Projection Screens
- └─ Video Conferencing (cameras)

AV Control & Switching

- |— Control Processors
- |— Touch Panels
- |— AV Switchers / Matrices
- |— Extenders (HDBaseT)
- |— Collaboration (ClickShare / wireless)
- └─ Signal Distribution

Unified Communications

- |— MTR Systems (Microsoft Teams Rooms)
- |— Zoom Rooms
- |— Video Bars
- └─ Speakerphones

Digital Signage

- |— Media Players
- |— CMS Licences
- └─ Mounts & Enclosures

Networking & Infrastructure

- |— Switches (PoE)
- |— Wireless Access Points
- |— Patch Panels
- └─ Racks & Enclosures

Cabling & Pathways

- |— CAT6 / CAT6A
- |— HDMI / AV Cable
- |— Audio Cable
- |— Speaker Cable
- |— Conduit & Trunking
- └─ Cable Management

Power

- └─ UPS
- └─ PDUs
- └─ Power Conditioning

Mounts & Structural

- └─ Display Mounts (fixed)
- └─ Display Mounts (adjustable)
- └─ Ceiling Mounts
- └─ Floor Stands
- └─ Rack Furniture

Consumables & Accessories

- └─ Faceplates & Wall Plates
- └─ Connectors & Terminations
- └─ Labels & Marking

Categories and subcategories are fully editable in the database admin panel.

4.3 Labour Database

Labour allocations are stored in the database and linked at the subcategory level. This allows consistent default hours that can be adjusted per quote.

Labour allocation record:

Field	Type	Notes
labour_id	string	
subcategory	string	FK to subcategory
task_description	string	e.g. "Install ceiling speaker, terminate, test"
hours_per_unit	decimal	Default install hours per product unit
labour_category	string	AV Tech / Electrician / Data Tech / PM
rate_per_hour	decimal	Loaded from Services rates
notes	string	Assumptions (e.g. "assumes accessible ceiling")

Labour hours auto-populate when a product is added to a space, based on subcategory

defaults. The estimator can override hours per line or per area.

4.4 Services Database

Services are non-product charge items. They are pre-built in the database and can be added to any quote, then manually adjusted.

Service record:

Field	Type	Notes
<code>service_id</code>	string	
<code>service_name</code>	string	e.g. "Project Management", "Commissioning", "Travel"
<code>category</code>	string	Labour / Expense / Subcontract
<code>unit</code>	string	hr / day / lot / % of project
<code>default_rate</code>	decimal	Default cost rate
<code>default_qty</code>	decimal	Default quantity (e.g. 1 lot)
<code>margin_pct</code>	decimal	Default margin for this service
<code>notes</code>	string	
<code>is_active</code>	boolean	

Common SKS services (pre-loaded):

- Project Management (hr or % of project)
- Site Supervision (day rate)
- Commissioning & Testing (hr)
- System Programming (hr)
- Documentation (lot)
- Training (hr)
- Travel & Accommodation (lot or expense)
- Freight & Delivery (lot)

- Cabling Labour — AV Tech (hr)
- Cabling Labour — Data Tech (hr)
- After-Hours Work (hr, premium rate)
- Subcontract — Electrical (lot / hr)
- Warranty & Support (lot)

4.5 Package / Bundle Library

Frequently combined products can be saved as packages in the database and applied to a space in one action.

Package record:

Field	Type	Notes
<code>package_id</code>	string	
<code>package_name</code>	string	e.g. "Standard Boardroom MTR System"
<code>description</code>	string	Short description
<code>category_tags</code>	array	For filtering
<code>items</code>	array	List of product_ids + default quantities
<code>default_labour</code>	array	Labour items included
<code>notes</code>	string	Assumptions
<code>created_by</code>	string	
<code>is_active</code>	boolean	

Packages are a starting point — all items remain individually editable once applied to a space.

5. Client & Project Information

5.1 Fields

Client Info:

- Client / Company Name
- Contact Name
- Contact Phone
- Contact Email
- Client Address

Project Info:

- Project Name
- Project Address / Site
- Quote Number (auto-generated: SKS-YYYY-XXXX)
- Quote Date (auto)
- Valid Until (auto: +30 days, adjustable)
- Revision (auto: Rev A, increments on re-issue)
- Prepared By (user logged in)
- Quote Type (Quick / Large)
- Internal Reference / Job Number
- Head Contractor (if applicable)
- Notes (internal)

5.2 AI-Assisted Population

From uploaded document (PDF, DOCX): User uploads a tender document, RFQ, or specification. Claude extracts:

- Client/company name
- Site/project name and address
- Contact details
- Any relevant project dates or deadlines

- Key scope notes (piped to the area builder or notes field)

From pasted email text: User pastes raw email content into a text box. Claude extracts the same fields.

Claude prompt (client extraction):

```
Extract the following information from the text below. Return as JSON only, no pr
Fields: client_name, contact_name, contact_email, contact_phone,
       project_name, site_address, project_notes.
If a field is not found, return null.

Text: [pasted email or document text]
```

6. Area / Space Builder (Large Quote)

6.1 Area Management

- User adds areas with the **[+ Add Area]** button
- Each area is named (e.g. "Level 3 Boardroom", "Reception", "Training Room A")
- Areas can be reordered via drag-and-drop
- Areas can be duplicated (useful for repeated spaces — e.g. "Meeting Room" × 4)
- Each area has:
 - Area name
 - Area type tag (optional — Boardroom / Meeting Room / Reception / etc.)
 - Internal notes / assumptions
 - Product lines
 - Labour summary
 - Area subtotal (cost and sell)

6.2 Product Selection Flow (per area)

```
[+ Add Product] button
|
├─ Filter by Category (dropdown)
```

- | └─ Audio / Visual / Control / UC / Signage / Network / Cabling / etc.
- |
- | └─ Filter by Subcategory (dropdown, dependent on Category)
- | └─ e.g. Ceiling Speakers / DSP / Displays / etc.
- |
- | └─ Filter by Brand / Manufacturer (optional dropdown)
- | └─ e.g. Biamp / Crestron / Samsung / Sony / etc.
- |
- | └─ Product Search (text search across description and part number)
- | └─ Results show: Brand | Model No. | Description | Unit | Sell Price (wi

On product selection:

- Line added to area with: Description, Model No., Qty (default 1), Unit, Cost Price, Margin %, Sell Price
- Labour hours auto-populated from subcategory default
- User adjusts quantity as required
- Sell price recalculates live

Line-level controls:

- Edit quantity (inline)
- Edit margin % (inline, overrides area/global default)
- Edit labour hours (inline)
- Add line note (internal only)
- Mark as Supply Only (zeroes labour)
- Mark as Provisional / PC Sum (flagged in output)
- Delete line
- Move line to another area

6.3 Package Application

- **[+ Add Package]** button opens package library
- User selects a package (e.g. "Standard Boardroom MTR")
- All package items are added to the area as individual editable lines
- Package name shown as a collapsible group header

- Items within the group can be expanded, edited, or removed individually
 - Multiple packages can be applied to one area
 - Custom groups can also be created manually (select lines → Group)
-

7. Margin Management

Margin is applied at multiple levels with a clear override hierarchy:

```
Global Default Margin (set in Settings)
└─ Category Override (e.g. Cabling at 40%)
    └─ Area Override (apply different margin to one space)
        └─ Line Override (individual product exception)
```

Margin controls:

- **Global margin** — set in the quote header bar (visible at all times)
- **Category margin** — set in the margin panel (expandable sidebar)
- **Per-line margin** — edit inline in the product table
- **Quick adjust** — “Apply X% to all lines” button resets all to a chosen margin
- Labour margin is managed separately (typically 0% on the labour rate itself)
- Services margin is set per service type in the Services section

Display toggle: User can switch the line-item view between:

- Sell price (default for quoting)
 - Cost price + margin (estimating view — never shown in client output)
-

8. Services Section

The Services section sits below the area product tables and applies to the whole quote (not per-area).

- Pre-loaded from the Services Database
- Services are toggled on/off with checkboxes

- Quantity and rate are editable inline
 - Additional services can be added manually (free-text description, qty, rate)
 - Labour hours summarised from product areas are displayed here for reference and reconciliation
 - Services are shown as a separate section in the client quote (or optionally included as a lump sum)
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9. Scope Document / Email Upload (Pre-population)

At any point before or during the quote build, the user can upload scope assistance documents:

Accepted inputs:

- PDF (drawings, specifications, scope of works, RFQ)
- Word document (tender docs, consultant specs)
- Pasted email text (copy-paste into text area)
- Image of a handwritten or printed scope (AI reads and interprets)

Claude extraction outputs:

- Suggested client/project details (piped to client info fields)
- Suggested product items matched against the product database
- Suggested area/space names from room schedules or drawings
- Unmatched scope items flagged for manual handling
- Notes and assumptions captured to the Notes field

User workflow:

1. Upload document → AI processes in background
 2. Review extraction results in a side panel
 3. Accept / reject / edit each suggestion individually
 4. Accepted items are added to the relevant area automatically
-

10. Review Screen

Before generating outputs, the user sees a full quote summary:

- All areas with item counts and area subtotals
- Services subtotal
- Cost breakdown (materials cost, labour cost, total cost)
- Margin breakdown (\$ and % by category)
- Sell price summary
- GST
- **Total (inc-GST)**
- Warnings:
 - Any unresolved unmatched items
 - Any lines missing quantities
 - Any areas with no products
 - Any services with \$0 rate

Editable from review screen:

- Global margin slider
- Quote validity date
- Notes to client (shown in output)
- Exclusions list (editable checklist)
- Terms (select from pre-loaded templates or edit inline)

11. Output Generation

11.1 Internal Cost Sheet (.xlsx)

Tabs:

Tab	Contents

Summary	Quote header, totals by area, margin summary, GST, grand total
Full BOM	All line items: Area / Category / Description / Model / Qty / Unit / Cost / Margin % / Sell / Labour Hrs
Services	All service lines with cost, margin, sell
Labour Detail	Hours by area, by trade, reconciliation against services
Settings	Margin defaults, labour rates (locked, reference only)

11.2 Client-Facing Quote (.docx)

Document structure:

Cover Page

SKS Logo

Quote Title: "Audio Visual Quotation"

Client Name & Address

Project Name

Quote No. | Date | Valid Until | Prepared By

Executive Scope Summary (AI-generated paragraph)

[LARGE QUOTE: Repeated per Area]

Area: [Area Name]

Item	Description	Model No.	Qty	Unit	Total
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Area Subtotal: \$x,xxx

Services & Installation

Project Management	\$x,xxx
Commissioning & Testing	\$x,xxx
...	

Pricing Summary

Materials	\$xx,xxx
Services & Installation	\$x,xxx

Subtotal (ex-GST)	\$xx,xxx
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GST (10%)	\$x,xxx
TOTAL (inc-GST)	\$xxx,xxx

Exclusions & Assumptions
[Checklist – editable before output]

Terms & Conditions
[Pre-loaded SKS standard terms]

SKS Footer: ABN | Address | Phone | Email | Web

Client quote rules:

- Only sell prices shown — cost price never exposed
- Model numbers: toggle show/hide
- Labour: toggle itemised per area OR lump sum in Services only
- Provisional / PC Sum items flagged with note
- Internal notes never shown
- Grouped items (packages) shown as collapsible sections or expanded (user selects)

12. Quote Management

Quotes are saved within the tool and can be managed over time.

Quote states:

Draft → Issued → [Accepted | Declined | Expired | Revised]

Quote list view shows:

- Quote number, client, project, value, date issued, status, prepared by

Actions on existing quotes:

- **Edit** — reopen in draft mode
- **Revise** — creates Rev B (keeps Rev A read-only)

- **Duplicate** — copy as new quote (new number, reset to Draft)
- **Convert to Job** — future: push to Simpro or export job data
- **Download** — re-download outputs without re-generating

Revision management:

- Each revision saves a snapshot of the full quote data
 - Revision history viewable (Rev A, Rev B, etc.)
 - Client document shows revision letter in header
-

13. Additional Capabilities (Recommended)

The following are worth including in the specification even if built in Phase 2:

Variations

- Ability to create a Variation quote linked to a parent quote/job
- Variation document format (e.g. "Variation No. 1 to Quote SKS-2026-0041")
- Delta pricing shown (additions / omissions)

Quote Templates

- Save a quote structure (areas + typical products) as a named template
- Apply template to new quote as a starting point
- e.g. "Standard 10-Person Boardroom", "Digital Signage Single Screen", "MTR Large Room"

Approval Workflow

- Quotes above a set threshold (e.g. \$50,000) require manager review before issue
- Approval status shown in quote list
- Approver can add comments or approve/reject

Contingency

- Optional contingency % field (added to totals, shown as separate line in internal sheet)

- Can be shown or hidden on client document

Alternate Options

- Ability to add an alternate product option to a line (e.g. Option A / Option B)
- Both options shown in client document with separate pricing
- Client selects preferred option

Price Validity Alerts

- Flag products where the price list is older than X days (configurable)
- Prompt user to check price before issue

Simpro Integration (Future)

- Export approved quote to Simpro as a new job/quote
- Map SKS product codes to Simpro catalogue IDs

Audit Log

- Record who created/edited/approved/issued each quote and when
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14. Database Admin Panel

A separate admin section for managing the underlying data:

Admin functions:

- Upload / update supplier price sheets
- Manage suppliers (add, edit, deactivate)
- Manage categories and subcategories (add, rename, reorder)
- Manage labour allocations (edit hours per subcategory)
- Manage services (add, edit, deactivate)
- Manage package library (create, edit, archive packages)
- Set global default margins per category
- Set labour rates per trade

- Manage quote templates
 - Manage exclusions library (standard exclusion clauses)
 - Manage terms & conditions templates
 - User management (add users, set roles: Admin / Estimator / Viewer)
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15. Tech Stack Recommendations

Layer	Recommended	Notes
Frontend	React + Tailwind	Component-based, good for complex forms and tables
Database	SQLite (local MVP) or PostgreSQL (hosted)	Relational DB suits the supplier/product/category structure
Backend API	Node.js + Express or Python FastAPI	Handles DB queries, file uploads, AI calls
AI Engine	Claude API (claude-sonnet-4-20250514)	Extraction, matching suggestions, scope summary
File parsing	pdf-parse / pdfminer, mammoth.js	PDF + DOCX text extraction
Excel output	SheetJS (xlsx)	Multi-tab cost sheet
Word output	docx (npm)	Client quote .docx
File storage	Local filesystem (MVP) / S3 (hosted)	Uploaded price sheets + generated outputs
Auth	Simple session auth (MVP)	Role-based in Phase 2

16. MVP Phase Plan

Phase 1 — Core Quoting (build first)

- Database schema and admin: supplier upload, product catalogue, categories
- Quote mode selection (Quick / Large)
- Client & project info form
- Area builder with product dropdown selection (Category > Subcategory > Brand > Product)
- Labour auto-allocation from subcategory defaults
- Services section
- Margin controls (global + per-line)
- Review screen with totals
- Excel cost sheet export
- Word client quote export

Phase 2 — Intelligence & Polish

- AI-assisted client info extraction (email/document upload)
- AI scope extraction → product suggestions from uploaded docs
- Package library and package application
- Quote save/load and quote list
- Revision management
- Quote templates

Phase 3 — Advanced

- Approval workflow
- Variations module
- Alternate options per line
- Simpro integration
- Price validity alerts
- Audit log
- Customer-facing quote portal link

17. Key Business Rules

- 1. **Client documents never expose cost price** — sell price only
- 2. **GST always calculated on sell price total** — shown as a separate line, never embedded
- 3. **Margin hierarchy:** Global → Category → Area → Line (lower level overrides higher)
- 4. **Labour margin = 0% by default** — the labour hourly rate already carries SKS margin
- 5. **Supply-only lines carry zero labour hours** automatically
- 6. **Provisional / PC Sum items** are flagged in both internal and client outputs
- 7. **Revision letters increment on re-issue** — never overwrite an issued quote
- 8. **Quote number format:** SKS-YYYY-XXXX (e.g. SKS-2026-0042)
- 9. **Validity period defaults to 30 days** from issue date (adjustable per quote)
- 10. **Packages are a starting point** — all items remain individually editable after application
- 11. **Price list import is additive + versioned** — previous cost prices are archived, not deleted

18. Claude API Prompt Summary

Stage	Purpose	Input	Output
Client extraction	Parse client/project info from email or doc	Raw text	JSON: client fields
Scope extraction	Identify products and areas from documents	Document text	JSON: suggested items + area names
Product matching	Match extracted items to product DB	Item descriptions + product list	Ranked match suggestions
Scope summary	Write client-facing intro paragraph	Approved line items	2–3 sentence professional summary
Exclusions suggestion	Suggest relevant exclusions for scope	Line items + area list	Suggested exclusion bullet points

All prompts must:

- Instruct Claude to return JSON only (no preamble) where structured output is needed
 - Be wrapped in try/catch with manual fallback on parse failure
 - Use model `claude-sonnet-4-20250514`
 - Keep max_tokens at 1000 for extraction tasks, 500 for summaries
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End of Framework Document v2.0